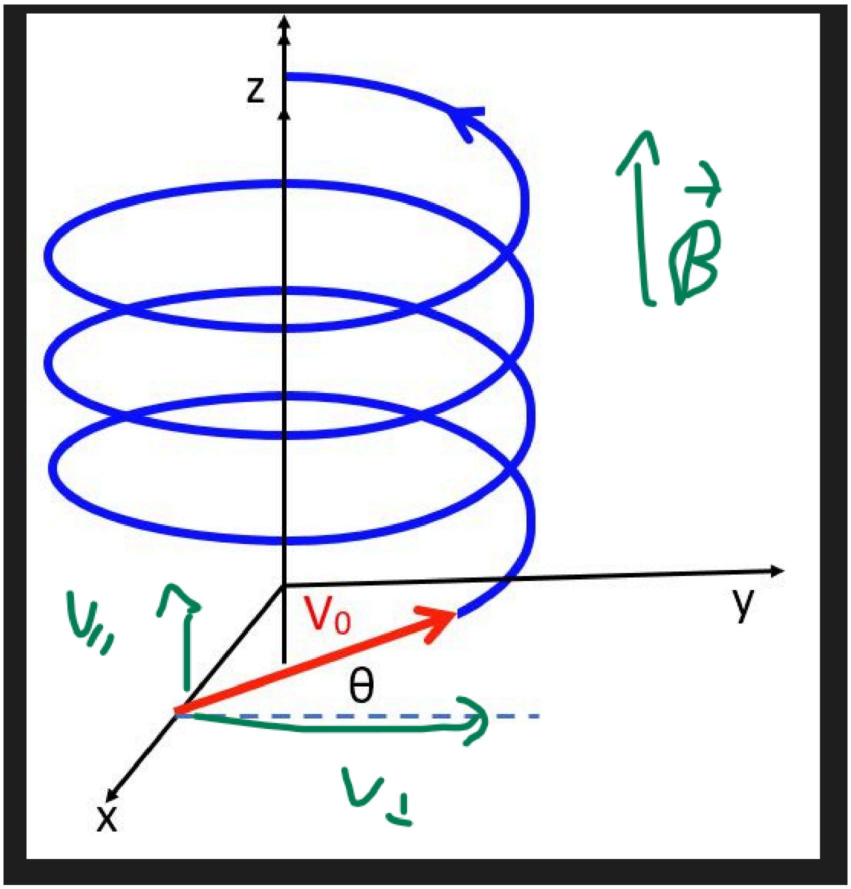
17_popquiz charge particle in a magentic field, electric field

This is a preview of the published version of the quiz

Started: Mar 22 at 7:14pm

Quiz Instructions



Question 1 12.5 pts

If a magnetic field is from up to down (vertical down). And if an electron enters this magnetic field with a velocity @ right. The force is

hint: its a negative charge. Use your left hand.

- away from you (in the screen)
- @ right

- ○ @ up
- ○ toward you (out of the screen)

Question 2 12.5 pts

positive isotopes enter a magnetic field B and have a circular motion (radius r). What is true.

The larger isotopes

- ○ All isotopes have the same radius
- ○ have the largest radius
- ○ I missed the class
- ○ have the smallest radius

Question 3 12.5 pts

positive isotopes enter a magnetic field B with a velocity V and have a circular motion (radius r). What is true.

- ○ if you increase B, the radius is larger. If you increase V the radius is smaller
- ○ if you increase B, the radius is larger. If you increase V the radius idoes not change
- ○ if you increase B, the radius is larger. If you increase V the radius is larger
- ○ if you increase B, the radius is smaller. If you increase V the radius is larger.

Question 4 12.5 pts

positive isotopes (charge q) enter a magnetic field B with a velocity V and have a circular motion (radius r). The time to complete a circle is T (see slides). what is true

- ○ T does not depend on m
- ○ T depends on m, V, B, q
- ○ T does not depend on V

- T does not depend on B
- T does not depend on q

Question 5 12.5 pts

A positive charge q and mass m enters an electric field E with a velocity Vx along the x-axis (@ right). The electric field is down (along negative y-axis).

The charge

- keeps going at a constant velocity along the x-axis. There is no component of the velocity Vy along the y-axis
- Stops moving
- The charge accelerates up (ay=qE/m) while moving @ right with the same Vx
- is accelerated down (ay=- qE/m) while moving at a constant velocity (Vx) along the x-axis
- is accelerated down (ay=- qE/m) but stops moving at right (Vx=0)

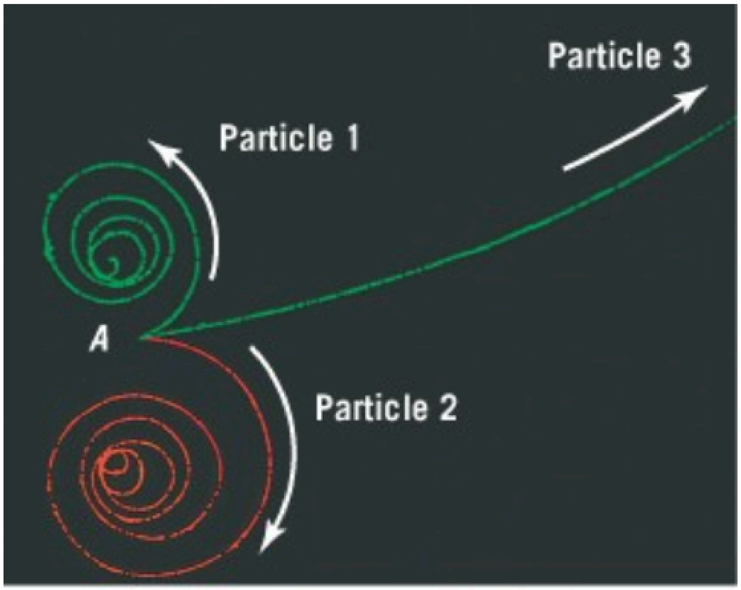
Question 6 12.5 pts

Building on the previous question. This motion describes

(remember throwing a stone in a gravitation field)

- a parabola curved down
- a vertical line
- a parabola curved up
- a horizontal line

Question 7 12.5 pts



particles enter a bubble chamber where a magentic field exists. The magnetic is in the screen. what is true

- particle 3 is positive
- particle 1 is positive.
- can not bedetermined
- particle 2 is positive

Question 8 12.5 pts

Building on previous question. the particles moving in the liquid spiral. The radius gets smaller. This is because

- its losing energy (momentum) so V decreases
- its losing mass so r decreases
- its losing energy (momentum) so B decreases
- its losing charge so r decreases

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